

Deepfake Detection Challenge Classroom Run Kit

TOK Cognitive Science Activity Bank • Knowledge and Technology • 25-35 min

Category	Knowledge and Technology	Time	25-35 min
Difficulty	Medium	ID	deepfake-detection-challenge

Big idea: Evidence standards change when media is synthetic.

One-Minute Teacher Brief

Students consider how synthetic media changes evidence standards.

Student Prompt

Inspect Deepfake Detection Challenge Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Deepfake Detection Challenge Stimulus A	Student-facing stimulus 1: Three image descriptions: authentic-looking, edited-looking, and ambiguous. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Deepfake Detection Challenge Stimulus B	Student-facing stimulus 2: A checklist for source, context, lighting, body geometry, and independent corroboration. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Use this for comparison, a second round, or a partner challenge.
Deepfake Detection Challenge Reveal / Twist	Student-facing stimulus 3: A debrief comparing technical clues with social trust cues. Rank, sort, or audit this output. Identify which rule, ranking signal, or interface choice shaped your judgement.	Teacher reveal: connect the changed response to the big idea — Evidence standards change when media is synthetic.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Deepfake Detection Challenge Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Deepfake Detection Challenge Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	What happens to knowledge when evidence can be easily fabricated?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Technology.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how evidence standards change when media is synthetic.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: What happens to knowledge when evidence can be easily fabricated?

Setup Checklist

- Prepare a starter stimulus using: Three image descriptions: authentic-looking, edited-looking, and ambiguous.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Example Stimuli or Materials

- Three image descriptions: authentic-looking, edited-looking, and ambiguous.
- A checklist for source, context, lighting, body geometry, and independent corroboration.
- A debrief comparing technical clues with social trust cues.

Resources To Use

- Resource pack: <resources/deepfake-detection-challenge/index.html>
- Card deck CSV: <resources/deepfake-detection-challenge/cards.csv>
- Visual prompt: <resources/deepfake-detection-challenge/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

Run Sequence

1. Show several media examples without revealing which are manipulated.
2. Students create a checklist: source, metadata, shadows, hands, audio mismatch, reverse image search, corroboration.
3. Reveal answers or uncertainty.
4. Students discuss how much trust should change in a synthetic media environment.

Debrief Moves

- Knowledge question: What happens to knowledge when evidence can be easily fabricated?
- Do new technologies require new standards of proof?
- Is "seeing is believing" still defensible?

Teacher Quick Script

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...